DATA 3402 Lab — Explore Your Kaggle Dataset

Getting from a downloaded file to data you actually understand

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Load → inspect → understand features → visualize → identify problems

The problem

Before modeling, we need to know what is actually in the data.

Today’s problem

- You have chosen a Kaggle dataset. Now you need to understand it before you model it.
- **By the end of lab, you should have:**
  - a table describing your features
  - a pair plot for a useful subset of numerical features
  - tables for categorical features
  - three things you learned about the data
  - one issue you need to fix before modeling
- If cleaning takes most of your time today, that is fine.

→ open your dataset and the lab notebook

First look: understand what you have

```
df = pd.read_csv("train.csv")

df.head()
df.shape
df.info()
df.isna().sum()
```

- What does one row represent?
- What are you trying to predict?
- Which columns have missing values?
- Do the pandas dtypes make sense?
- Are there obvious ID columns or duplicate rows?

The first goal is not to “fix” anything. It is to understand what you have.

→ notebook: load + first look

Build a feature table

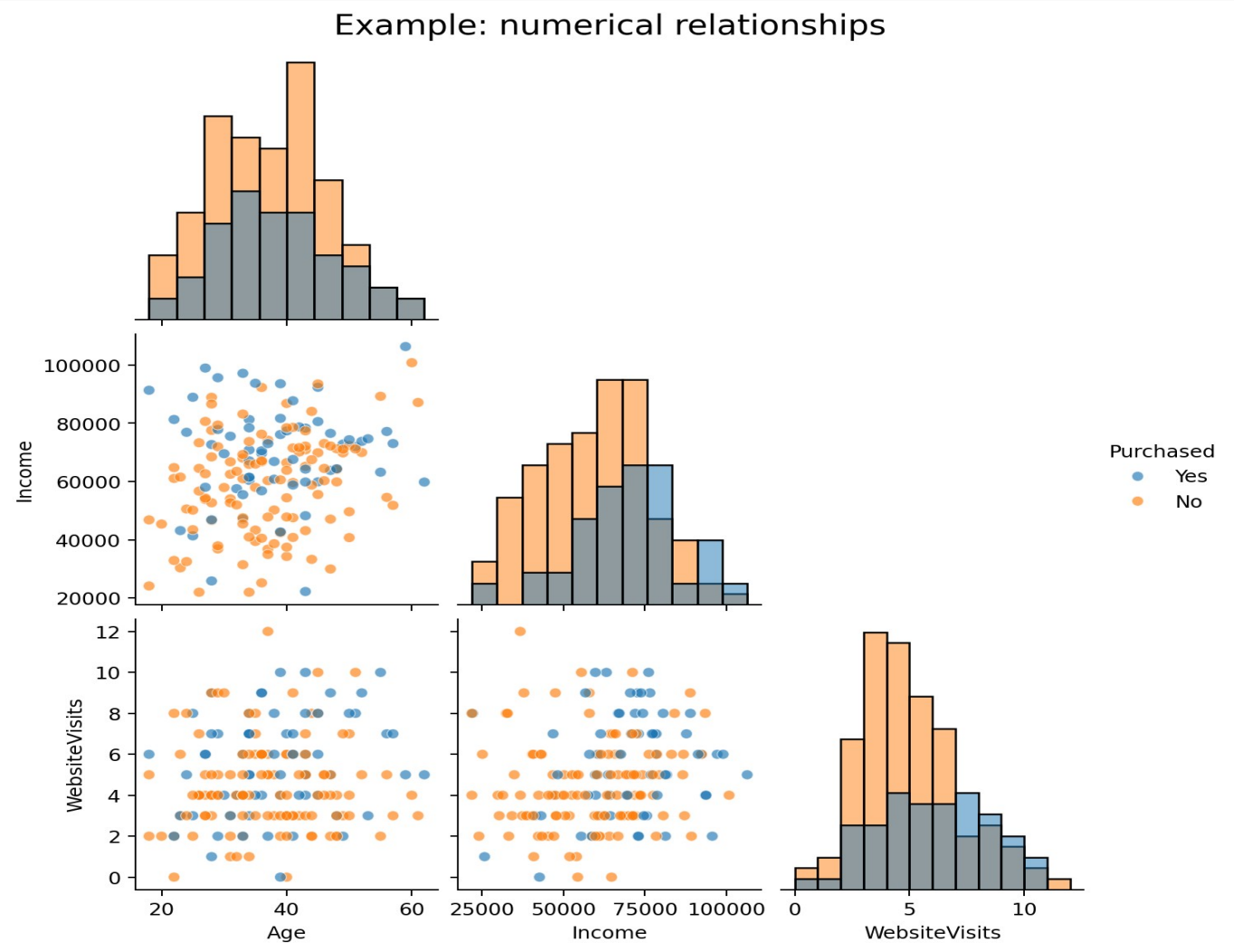
For each feature — or sensible feature group in a very wide dataset — record what the data actually means.

Feature	Meaning	Role	Values / range	Missing	Outlier notes
Age	customer age	numerical	18–72	0	check tails
Gender	customer gender	categorical	Female / Male	5	n/a
Income	annual income	numerical	$22k–$140k	7	review extremes
Region	sales region	categorical	N / S / E / W	3	n/a
Purchased	target	target	Yes / No	0	n/a

- **Storage type ≠ meaning**
- An integer can still represent categories.
- An object column may be free text, an ID, or a true category.
- For very wide datasets, describe groups of related features.

→ notebook: feature summary table

Numerical features: pair plots



- **Choose 4–6 meaningful numerical features.**
- Look for relationships between variables.
- Look for class separation if you have a categorical target.
- Look for clusters, outliers, or redundant features.
- **Do not pair-plot 30 columns just because you can.**

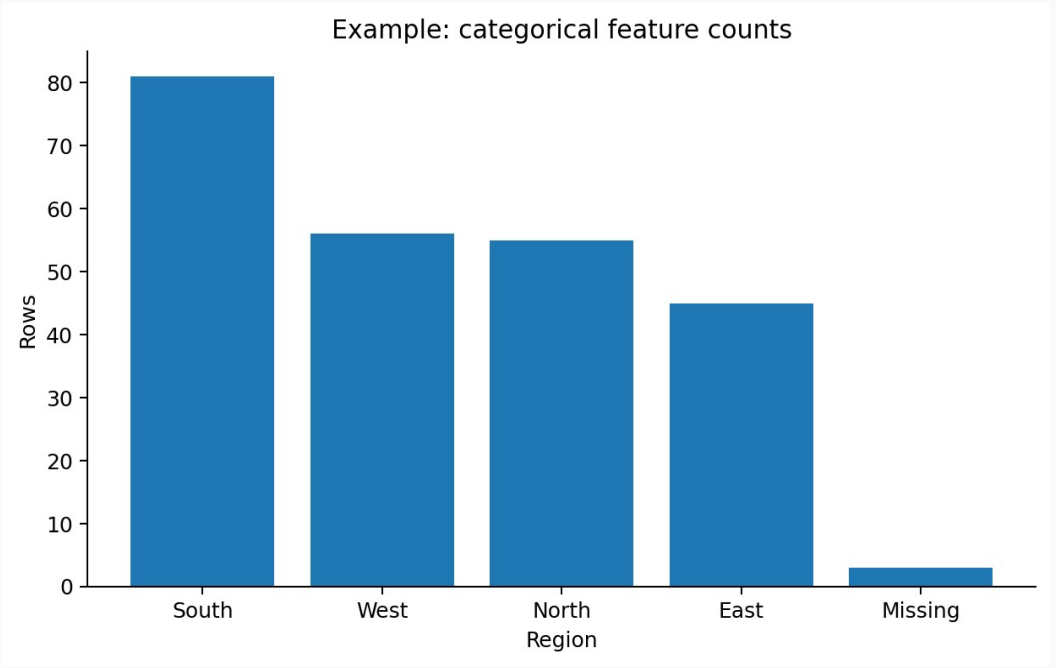
→ notebook: numerical exploration

Categorical features: make tables instead

```
df["Region"].value_counts()

df["Region"].value_counts(normalize=True)

pd.crosstab(df["Region"], df["Target"],
            normalize="index")
```



- Counts tell you how common each category is.
- Percentages make categories easier to compare.
- A cross-tab lets you ask whether the target behaves differently across categories.
- Very high cardinality may mean an ID, free text, or a feature that needs special handling.

→ notebook: categorical exploration

When it breaks: cleaning is part of the job

Things you may discover today:

- missing values or blank strings
- numbers stored as text; dates stored as strings
- inconsistent labels (Male / male / M)
- ID columns that probably should not be model inputs
- outliers that may be errors — or may be real
- one target class that dominates the dataset

Do not “fix” something automatically just to make the code run.

First understand what the problem is. Then decide what treatment makes sense for your dataset.

Cleaning is not a detour from the project. It is part of the project.

Your turn

Work through your own Kaggle data. Ask for help when you hit a blocker.

Today’s checkpoint

- Dataset loads successfully.
- Target and class balance are checked.
- Rows, columns, dtypes, and missing values are checked.
- Feature summary table is created and reviewed.
- A useful subset of numerical features is explored with a pair plot.
- Categorical features are summarized with tables.
- Write three things you learned about the dataset.
- Write one issue you need to fix before modeling.

If you finish early: investigate one suspicious feature or one cleaning decision — do not rush straight to a complicated model.

Before you leave

Be able to answer four questions about your dataset:

- **What is your target?**
- **What are three things you learned about the data?**
- **What is one problem you need to fix before modeling?**
- **What is your next concrete step?**

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understand → clean / preprocess → baseline model → evaluate → improve

The goal today was not the highest Kaggle score. It was to understand what you are feeding into the model.